

Orthogonal Calibration in ETFT

Preventing Trajectory Overfitting via Specialized Depth-Width Multi-Agent Fine-Tuning

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Abstract

A critical vulnerability in the Evolutionary Trajectory Fine-Tuning (ETFT) framework is the risk of "Trajectory Overfitting"—the model memorizing a single historical path ($a_0 \rightarrow a_n$) rather than learning the generalized mechanics of innovation. To ensure the model acts as a creative architect rather than a stochastic parrot, we introduce **Orthogonal Calibration**. This strategy splits the fine-tuning process into two distinct axes: **Depth Models** (trained on the full chronological sequence to understand the macro-vector of innovation) and **Width Models** (narrowly specialized on specific generations to maximize lateral creativity and architectural variance). This multi-agent ensemble allows for recursive self-improvement and objective validation of the SOTA+1 capability.

1 The Calibration Blindspot: Overfitting the Vector

If a standard code-generation LLM is fine-tuned monolithically on the sequential trajectory $\mathcal{T} = (a_0, a_1, \dots, a_n)$, the optimization landscape forces the model to minimize the loss against the specific human-designed jumps. The model learns *what* humanity did, but not necessarily *how else* it could be done.

When tasked with generating a_{n+1} , a monolithic model often outputs a highly conservative, mathematically safe perturbation of a_n . We require an architecture that balances the structural soundness of historical progression with the lateral creativity required for paradigm-shifting leaps.

2 The Orthogonal Fine-Tuning Strategy

We propose splitting the generative responsibility into an ensemble of specialized models, calibrated along two orthogonal axes: **Depth** and **Width**.

2.1 1. The Depth Model (The Historian)

- **Training Scope:** The entire chronological trajectory ($a_0 \rightarrow a_n$) across a narrow set of domains.
- **Objective Function:** Maximizing the understanding of the *macro-vector* of innovation. It learns the "story" of how brute force transforms into elegant sparsity.

- **Role in ETFT:** It provides the structural context and prevents the overall system from suggesting architectures that mathematically regress into past paradigms.

2.2 2. The Width Models (The Generation Specialists)

- **Training Scope:** Narrowly specialized on a single evolutionary jump (e.g., $M_{width}^{n-1 \rightarrow n}$ is trained *only* on innovating from a_{n-1} to various forms of a_n).
- **Data Augmentation (Lateral Training):** For a specific jump (a_{n-1}), the dataset contains not one, but multiple, architecturally diverse, objectively functional solutions that achieve the a_n performance threshold.
- **Objective Function:** Maximizing architectural variance (creativity) while meeting strict objective performance metrics (FLOPs, Memory).

3 Objective Calibration and SOTA+2 Recursion

Objective Calibration (Proving Gen+1 Capability):

The efficacy of a Width Specialist can be objectively measured. We feed the model a_{n-1} and task it to generate novel architectures. If the model outputs a solution that is architecturally distinct from the known historical a_n , yet achieves equivalent or superior physical metrics (verified by the in-silico sandbox), the model is objectively calibrated as a "Creative Innovator."

Recursive Synthesis (The SOTA+2 Extrapolation):

Because the Width Models are trained to produce architecturally diverse jumps rather than sequential memory, we can enable recursive synthesis. By taking the newly generated SOTA+1 architecture and feeding it back into the SOTA-calibrated Width Model as a new input, the model can theoretically synthesize the **SOTA+2** architecture—extrapolating multiple generations into the future before human engineers have even conceptualized SOTA+1.

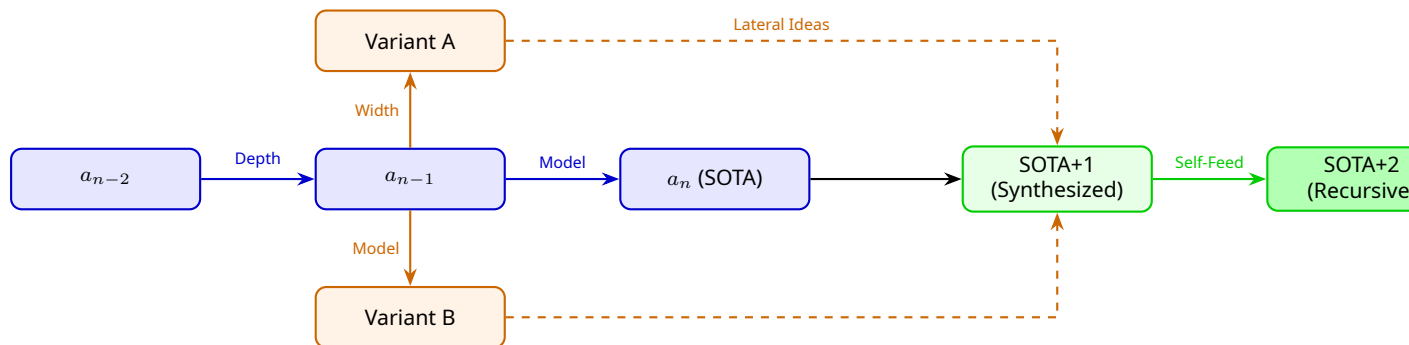


Figure 1: The Orthogonal Calibration Architecture. The Depth Model ensures historical continuity, while Width Models train on architectural variance at specific generations. This combination prevents trajectory overfitting, enables creative lateral innovation, and theoretically allows for recursive SOTA+2 generation.

4 Conclusion

By splitting the fine-tuning process into Depth (historical vector) and Width (generational variance) axes, we systematically prevent the ETFT system from devolving into a static historical repeater. This Orthogonal Calibration strategy ensures the synthesis of creative, architecturally diverse solutions that can be objectively validated against performance metrics. Furthermore, it unlocks the profound capability of recursive self-feeding, theoretically enabling the system to jump multiple generations (SOTA+2) ahead of current human scientific limits.